

Breast Reconstruction Timing Algorithm

Who gets chemotherapy and when, who gets radiation, what that does to the reconstruction, and how long to wait at each step. Oncology makes the final call; this is what you should be able to predict and counsel from before the tumor board note arrives.

Master algorithm: subtype to reconstruction plan

Subtype decides chemotherapy and its order. Nodes and tumor size decide radiation. Radiation decides the reconstruction. Read top to bottom.

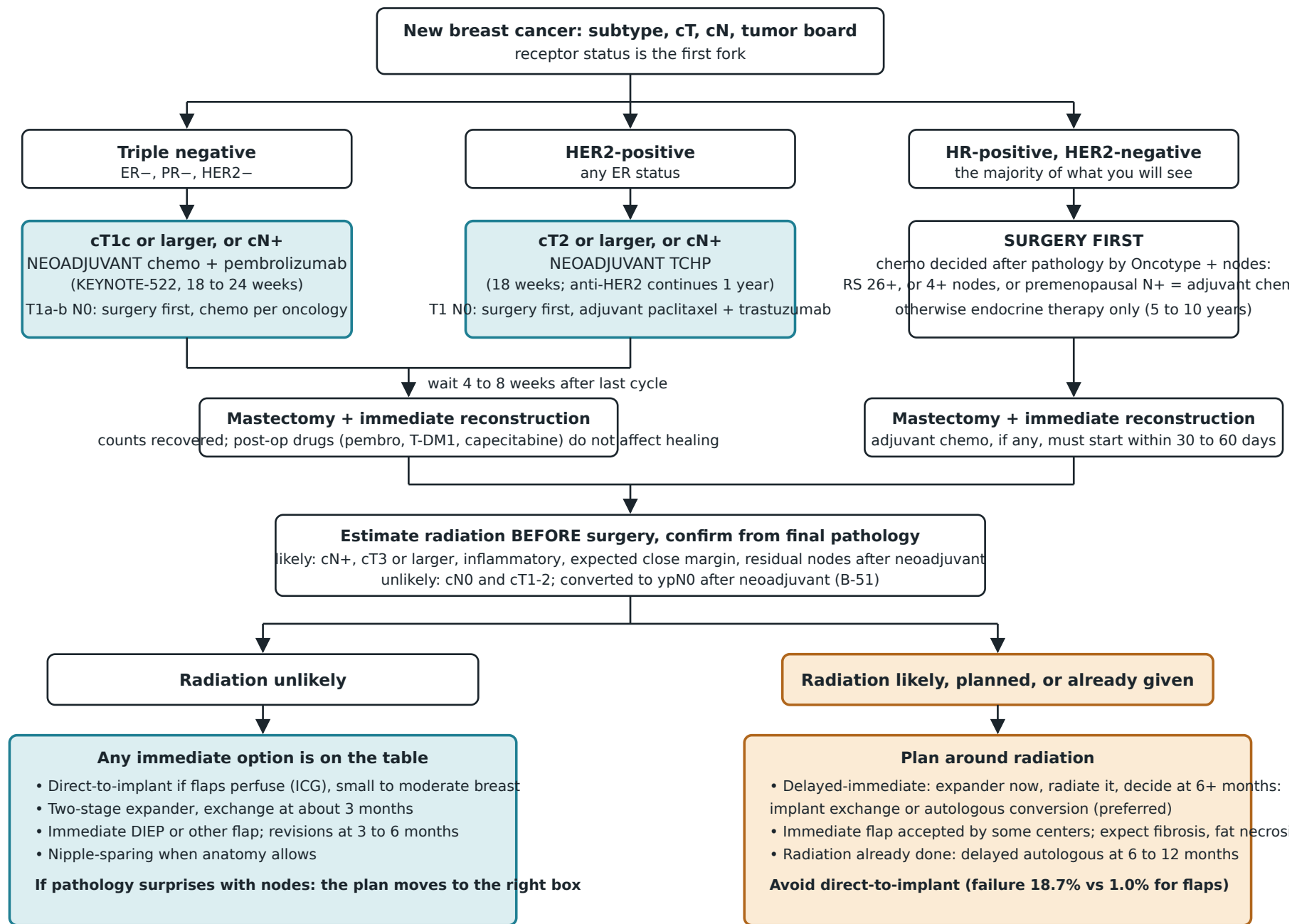


Figure 1. Triple-negative and HER2-positive disease above the smallest tumors gets chemotherapy before surgery; hormone-receptor-positive disease usually gets surgery first and chemotherapy only if genomics or nodes demand it. Radiation likelihood, estimated before surgery and confirmed by pathology, decides between the two reconstruction boxes at the bottom.

Calendar timelines for the four pathways

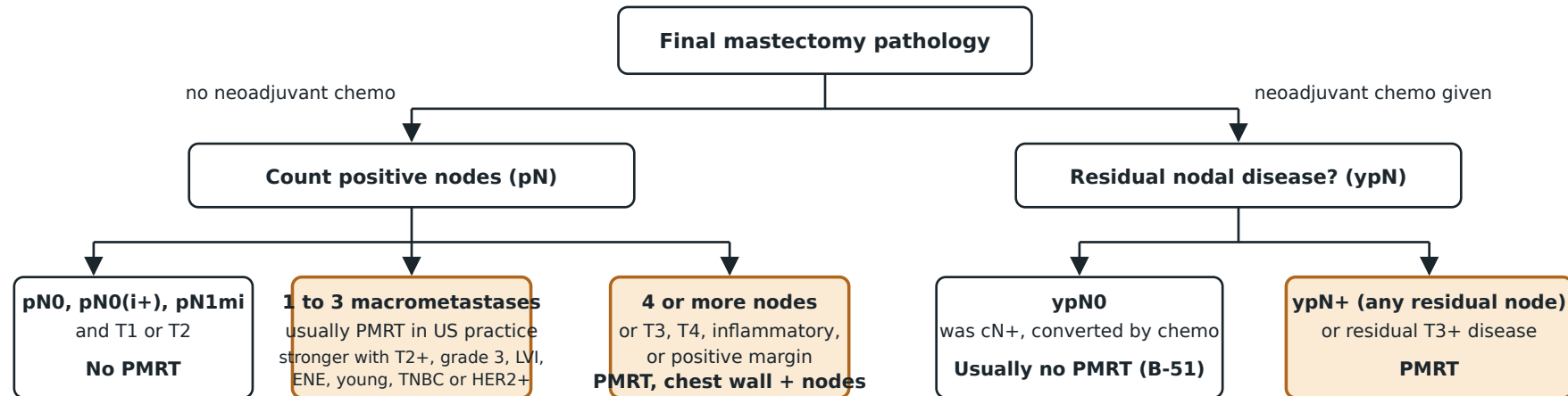
Weeks from the anchoring event. The shaded blocks are the periods when you should not operate electively; the marked points are the earliest dates for your next stage.



Figure 2. Four pathways on a common week scale. In A and B the expander or flap is radiated and nothing elective happens for six months afterward; in C the whole reconstruction finishes inside six months; in D the reconstruction waits out the acute phase and the three-to-six-month complication peak.

Radiation decision after mastectomy

This is the tree oncology walks. Knowing it lets you predict the plan the day the pathology posts.



2025): no survival benefit of chest wall RT in pN1; practice may loosen, assume PMRT until told otherwise

The axilla: a positive sentinel node after mastectomy is often treated with axillary radiation instead of completion dissection (AMAROS)

So "positive node" almost always means radiation is coming. Ask whether dissection is planned: dissection plus radiation is the highest lymphedema risk.

Timing: no chemo, RT starts 3 to 6 weeks after surgery. With adjuvant chemo, RT follows chemo, about 4 to 6 months after mastectomy.

Figure 3. Without neoadjuvant chemotherapy the node count decides; after neoadjuvant chemotherapy any residual node decides. The shaded boxes are the outcomes where your reconstruction must be planned around radiation.

Reconstruction options against radiation status

	Radiation unlikely cN0, T1-2, or ypN0	Radiation possible or planned cN+, T3, ypN+, or awaiting pathology	Radiation already given delayed reconstruction
Direct-to-implant one stage, ICG-dependent	Recommended if flaps perfuse small to moderate, minimal ptosis	Avoid failure 18.7%, contracture 15 to 50%	Avoid irradiated envelope will not stretch
Expander then implant two stages	Recommended exchange at about 3 months	Acceptable with counseling radiate expander, exchange at 6+ months	Acceptable, higher failure often needs latissimus or fat grafting
Immediate autologous DIEP, PAP, LD	Recommended best long-term satisfaction	Acceptable with counseling fibrosis, 10 to 20% volume loss, fat necrosis	not applicable
Delayed-immediate expander now, decide later	Acceptable when pathology could still surprise	Recommended expander now; flap or implant at 6+ months	not applicable
Delayed autologous after all cancer treatment	Acceptable patient preference or medical delay	Acceptable flat closure now, flap after RT	Recommended 6 to 12 months after RT; failure 1%

 recommended
 acceptable with counseling
 avoid

Figure 4. The one cell that changes everything is direct-to-implant under radiation: recommended on the left, avoid on the right. Delayed-immediate expansion is the hedge when radiation is possible; delayed autologous reconstruction is the answer once radiation has happened.

Waiting intervals to write in the plan

AFTER THIS	WAIT THIS LONG BEFORE	THEN	WHY
Last neoadjuvant chemo cycle	4 to 8 wk	Mastectomy and immediate reconstruction	Count recovery; beyond 8 weeks survival worsens
Mastectomy (adjuvant chemo planned)	≤ 30 to 60 d	Chemo must start; your wound must be closed and dry	Delay past 60 days increases mortality, most in TNBC and HER2+
Mastectomy (radiation, no chemo)	3 to 6 wk	Radiation starts; expansion complete before simulation	Local control; volume is fixed at simulation
Last chemo cycle (radiation to follow)	3 to 4 wk	Radiation starts	Marrow recovery; taxane spacing
Radiation ends (expander in place)	≥ 6 mo	Exchange to implant or conversion to flap	Failure 22.4% under 6 months vs 7.7% after
Radiation ends (no reconstruction)	6 to 12 mo	Delayed autologous reconstruction	Complications peak at 3 to 6 months; skin must be supple
Radiation ends (flap already in place)	≥ 6 mo	Revisions, fat grafting, nipple reconstruction	Fibrosis and volume loss declare by 6 to 12 months
Radiation ends (any reconstruction)	9 to 12 mo	Contralateral symmetry	Radiated side must stabilize first
Fat grafting session in irradiated tissue	~3 mo	Next session or expander exchange	Graft take and skin conditioning

AFTER THIS	WAIT THIS LONG BEFORE	THEN	WHY
Bevacizumab, last dose	≥ 28 d	Any elective surgery; restart ≥ 28 d after and wound healed	FDA label; half-life 21 days
Tamoxifen (free flap planned)	28 d hold	Microsurgery, as a shared decision with oncology	Kelley 2012 flap complications vs 2022 meta-analysis
Nicotine, any form	4 wk before and after	Any reconstruction	ASPS position
Expander fills during chemo	avoid days 7 to 14 of each cycle	Fill between cycles	Neutropenic nadir

Pembrolizumab, trastuzumab, pertuzumab, T-DM1, capecitabine, and aromatase inhibitors need no hold and do not impair healing. Screen checkpoint-inhibitor patients for thyroid and adrenal effects before anesthesia.

What to say in each scenario

Triple negative, node negative, chemo done

"Your chemotherapy is finished. We operate four to eight weeks after your last cycle. Radiation is unlikely if the nodes stay clear, so any reconstruction is open to you. If the pathology finds a node, we will have placed an expander that can be radiated and finish the reconstruction six months after radiation."

HER2-positive, node positive at diagnosis

"After your chemotherapy and mastectomy, radiation depends on whether any cancer remains in the nodes. If they are clear, probably no radiation. If not, radiation is expected. Because of that uncertainty I recommend an expander now and the definitive reconstruction, implant or your own tissue, six months after radiation. Your Herceptin continues through all of this without affecting surgery."

Hormone-positive, postmenopausal, clinically node negative

"You will most likely not need chemotherapy or radiation; the final decision waits on the Oncotype test and the nodes. This is the situation where a one-stage implant or an immediate flap is most reasonable. If a node surprises us, the plan changes and I will tell you how."

**Hormone-positive,
premenopausal, node positive**

"Chemotherapy after surgery is likely and radiation is probable. Chemotherapy needs to start within about six weeks, so the first job of the reconstruction is to heal cleanly. I recommend an expander now, filled between chemo cycles, radiated, and converted to an implant or your own tissue six months after radiation."

Inflammatory or T4, or four or more nodes

"Radiation is certain. I recommend no implant at the time of mastectomy. Either an expander to hold the skin envelope, or a flat closure with a tissue flap reconstruction six to twelve months after radiation. Your own tissue tolerates radiation; implants fail about one in five times in this setting."

Delayed patient, radiation finished

"The best window for reconstruction with your own tissue is six to twelve months after radiation, once the skin has softened. If we use an expander instead, expect higher failure and probably a latissimus flap or fat grafting to help the skin. Contralateral matching waits until the reconstructed side has settled."

The one sentence to put in every note: "Adjuvant plan anticipated: [chemo yes/no, timing], [radiation likely/unlikely, basis]; reconstruction sequenced accordingly with [expander / flap / implant] and definitive stage no earlier than [date]." That sentence is Diagnosis/Planning and Safety in one line.